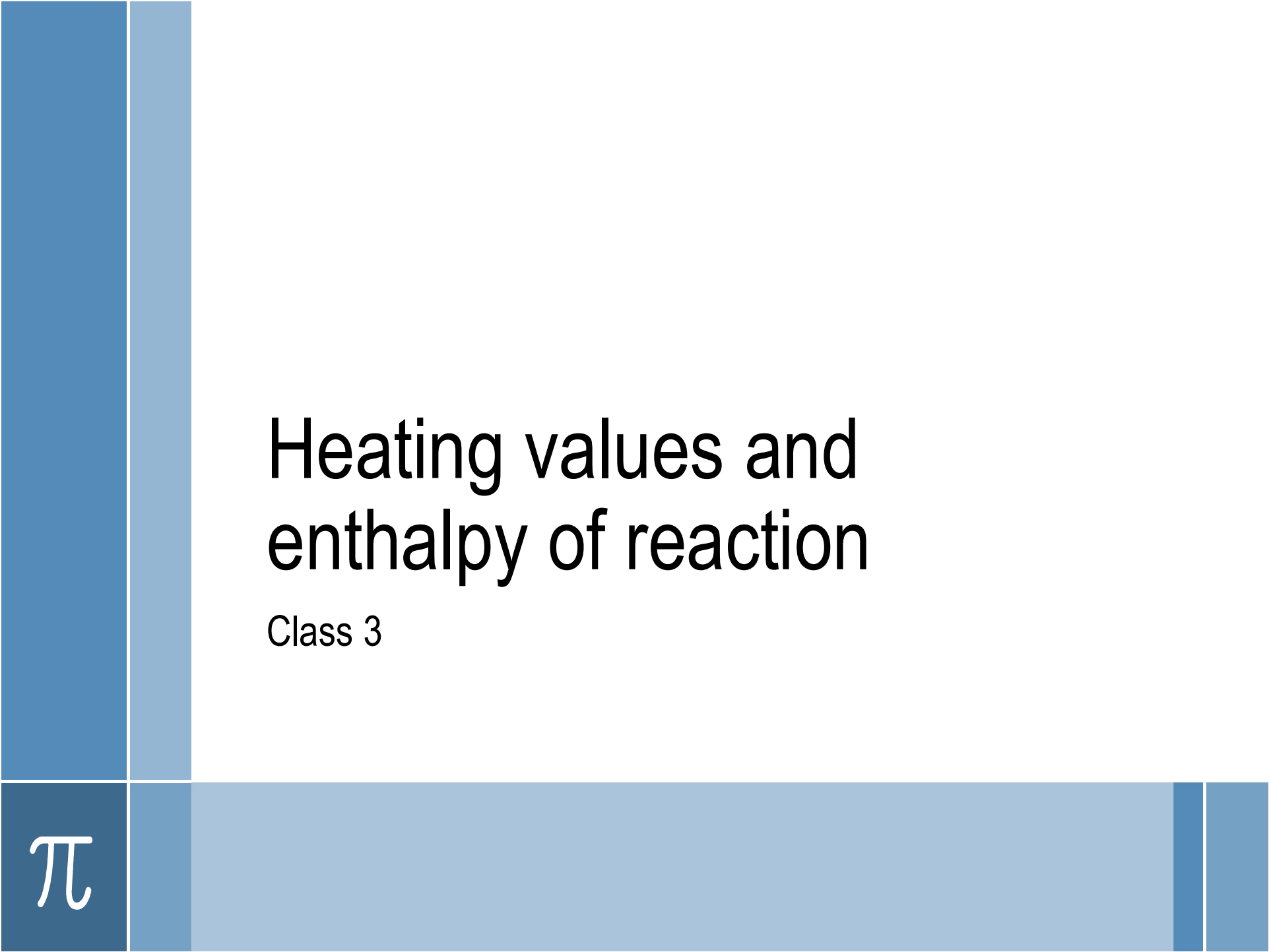




Heating values and enthalpy of reaction

Class 3

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Objective

- › Definition of enthalpy of formation
- › Definition of absolute and sensible enthalpy
- › Heating values and enthalpy of reaction
- › Energy balances for reactive systems
 - Open and close systems

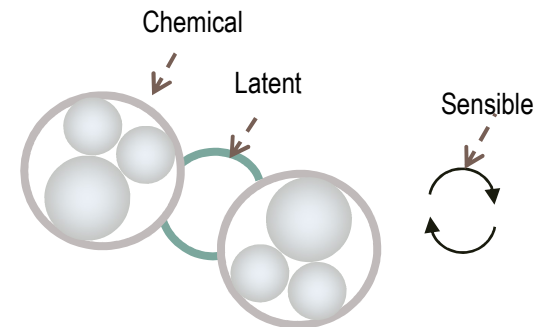
Enthalpy and reference state

- › Enthalpy (H, h: joule).
 - Property of the system.
 - We will use it to measure the internal energy.
- › Reference state
 - A state where a property can have a value of zero
 - › Gives us a reference point to our analysis
- › Standard reference state
 - **T = 25 °C and p = 1 atm.**
 - **It will be marked/denoted by the symbol (°)**
 - › h° – enthalpy at standard state
 - › u° – internal E at standard state

$$\Delta E_{\text{syst}} = \Delta E_K + \Delta E_P + \Delta U$$

$$\Delta E_{\text{syst}} = \Delta U$$

$$\Delta E_{\text{syst}} = \Delta U = \Delta H$$



Enthalpy of formation

- › Enthalpy of formation ($\bar{h}_f^\circ$)
 - Energy absorbed/liberated to obtain one kmol or kg formed from its elements,
 - When the compound and elements are at reference state.
- › Stable compounds
 - Certain elements will have an enthalpy value of zero
 - Those that at standard ref state are stable compounds
 - › $\text{H}_2, \text{O}_2, \text{N}_2$
 - There are also elements that exists in different states
 - › Carbon exists as graphite and diamond
- › Associated sign:
 - Follow the convention for heat transfer
 - › From compounds to reactor (+) endothermic reaction.

ENTHALPY OF FORMATION

Data for enthalpy formation at standard state.

g – gas

l – liquid

s - solid

Sustancia	Fórmula	$\bar{h}_f^\circ$ kJ/kmol
Acetileno	$C_2H_2(g)$	+226,730
Agua	$H_2O(l)$	-285,830
Alcohol etílico	$C_2H_5OH(g)$	-235,310
Alcohol etílico	$C_2H_5OH(l)$	-277,690
Alcohol metílico	$CH_3OH(g)$	-200,670
Alcohol metílico	$CH_3OH(l)$	-238,660
Amoniaco	$NH_3(g)$	-46,190
Benceno	$C_6H_6(g)$	+82,930
n-butano	$C_4H_{10}(g)$	-126,150
Carbón	$C(s)$	0
Dióxido de carbono	$CO_2(g)$	-393,520
n-dodecano	$C_{12}H_{26}(g)$	-291,010
Etano	$C_2H_6(g)$	-84,680
Etileno	$C_2H_4(g)$	+52,280
Hidrógeno	$H_2(g)$	0
Hidrógeno	$H(g)$	+218,000
Hidroxilo	$OH(g)$	+39,460
Metano	$CH_4(g)$	-74,850
Monóxido de carbono	$CO(g)$	-110,530
Nitrógeno	$N_2(g)$	0
Nitrógeno	$N(g)$	+472,650
n-octano	$C_8H_{18}(g)$	-208,450
n-octano	$C_8H_{18}(l)$	-249,950
Oxígeno	$O_2(g)$	0
Oxígeno	$O(g)$	+249,190
Peróxido de hidrógeno	$H_2O_2(g)$	-136,310
Propano	$C_3H_8(g)$	-103,850
Propileno	$C_3H_6(g)$	+20,410
Vapor de agua	$H_2O(g)$	-241,820

Absolute enthalpy

- › Term to measure the enthalpy of a compound at a state different from standard state.
- › Will be the sum of:
 - Enthalpy of formation
 - › Creating the compound.
 - Sensible enthalpy:
 - › Change between the standard and the state of interest.
 - Change of state at constant composition.

Absolute enthalpy
of i @ T, p

$$\bar{h}_i(T, p) = \bar{h}_{f,i}^\circ + \Delta\bar{h}_{s,i}(T, p)$$

Of formation i
@ T, p

Of formation I
@ T, p

$$\Delta\bar{h}_{s,i}(T, p) = \bar{h}(T, p) - \bar{h}(T_{ref}, p_{ref})$$

Sensible enthalpy
of i @ T, p

There are tables with the enthalpy values at different temperatures and constant pressure.

Propiedades de gas ideal del oxígeno, O₂

T K	$\bar{h}$ kJ/kmol	$\bar{u}$ kJ/kmol	$\bar{s}^\circ$ kJ/kmol · K	T K	$\bar{h}$ kJ/kmol	$\bar{u}$ kJ/kmol	$\bar{s}^\circ$ kJ/kmol · K
0	0	0	0	600	17,929	12,940	226.346
220	6,404	4,575	196.171	610	18,250	13,178	226.877
230	6,694	4,782	197.461	620	18,572	13,417	227.400
240	6,984	4,989	198.696	630	18,895	13,657	227.918
250	7,275	5,197	199.885	640	19,219	13,898	228.429
260	7,566	5,405	201.027	650	19,544	14,140	228.932
270	7,858	5,613	202.128	660	19,870	14,383	229.430
280	8,150	5,822	203.191	670	20,197	14,626	229.920
290	8,443	6,032	204.218	680	20,524	14,871	230.405
298	8,682	6,203	205.033	690	20,854	15,116	230.885
300	8,736	6,242	205.213	700	21,184	15,364	231.358
310	9,030	6,453	206.177	710	21,514	15,611	231.827
320	9,325	6,664	207.112	720	21,845	15,859	232.291
330	9,620	6,877	208.020	730	22,177	16,107	232.748
340	9,916	7,090	208.904	740	22,510	16,357	233.201
350	10,213	7,303	209.765	750	22,844	16,607	233.649
360	10,511	7,518	210.604	760	23,178	16,859	234.091
370	10,809	7,733	211.423	770	23,513	17,111	234.528
380	11,109	7,949	212.222	780	23,850	17,364	234.960
390	11,409	8,166	213.002	790	24,186	17,618	235.387
400	11,711	8,384	213.765	800	24,523	17,872	235.810
410	12,012	8,603	214.510	810	24,861	18,126	236.230
420	12,314	8,822	215.241	820	25,199	18,382	236.644
430	12,618	9,043	215.955	830	25,537	18,637	237.055
440	12,923	9,264	216.656	840	25,877	18,893	237.462
450	13,228	9,487	217.342	850	26,218	19,150	237.864
460	13,525	9,710	218.016	860	26,559	19,408	238.264
470	13,842	9,935	218.676	870	26,899	19,666	238.660
480	14,151	10,160	219.326	880	27,242	19,925	239.051
490	14,460	10,386	219.963	890	27,584	20,185	239.439
500	14,770	10,614	220.589	900	27,928	20,445	239.823
510	15,082	10,842	221.206	910	28,272	20,706	240.203
520	15,395	11,071	221.812	920	28,616	20,967	240.580
530	15,708	11,301	222.409	930	28,960	21,228	240.953
540	16,022	11,533	222.997	940	29,306	21,491	241.323
550	16,338	11,765	223.576	950	29,652	21,754	241.689
560	16,654	11,998	224.146	960	29,999	22,017	242.052
570	16,971	12,232	224.708	970	30,345	22,280	242.411
580	17,290	12,467	225.262	980	30,692	22,544	242.768
590	17,609	12,703	225.808	990	31,041	22,809	242.120

Reaction enthalpy (h_{rxn})

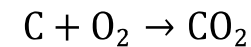
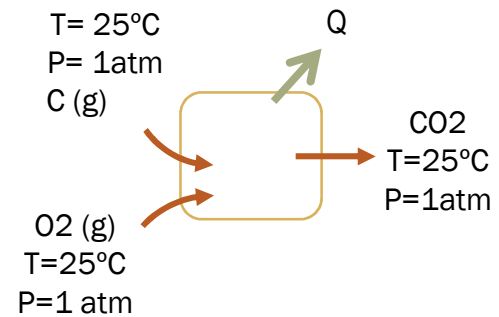
- › The difference between the enthalpy of products at a given state and the reactants at the same state, for a complete reaction:

- › The energy released or absorbed during the reaction.

$$\bar{h}_{\text{rxn}} = \sum_{\text{prod}} N_p \bar{h}_p - \sum_{\text{react}} N_r \bar{h}_r$$

- › Enthalpy of combustion (h_c):

- Term used instead of reaction enthalpy.



- Constant pressure and temperature
- At reference state

$$\begin{aligned} \bar{h}_{\text{rxn}} &= N_{\text{CO}_2} \bar{h}_{\text{CO}_2} - (N_{\text{C}} \bar{h}_{\text{C}} + N_{\text{O}_2} \bar{h}_{\text{O}_2}) \\ \bar{h}_{\text{rxn}} &= N_{\text{CO}_2} (\bar{h}_f^\circ + \bar{h}_{298} - \bar{h}_{298})_{\text{CO}_2} - N_{\text{C}} (\bar{h}_f^\circ + \bar{h}_{298} - \bar{h}_{298})_{\text{C}} \\ &\quad - N_{\text{O}_2} (\bar{h}_f^\circ + \bar{h}_{298} - \bar{h}_{298})_{\text{O}_2} \end{aligned}$$

$$\begin{aligned} \bar{h}_{\text{rxn}} &= (1\text{kmol})(-393520 + 9364 - 9364) \left(\frac{\text{kJ}}{\text{kmol}} \right) \\ &\quad - (1\text{kmol})(0 + 5690 - 5690) \left(\frac{\text{kJ}}{\text{kmol}} \right) \\ &\quad - (1\text{kmol})(0 + 8682 - 8682) \left(\frac{\text{kJ}}{\text{kmol}} \right) \\ \bar{h}_{\text{rxn}} &= \bar{h}_{\text{C}} = -393\,520 \frac{\text{kJ}}{\text{kmol fuel}} \end{aligned}$$

Heating value (HV)

- › Heat transferred from the chamber during combustion or reaction at constant pressure.
 - When the fuel is burned completely
 - When reactants and products are in the same state
- › Higher/lower heating value
 - Important the state of the water present in products
 - Higher heating value (HHV): water is liquid.
 - Lower heating value (LHV): vapour water.

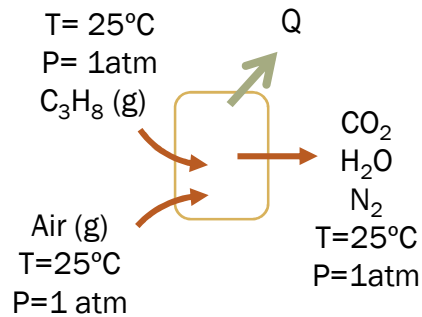
$$HV = |h_c| = \left(\frac{\text{kJ}}{\text{kg}_{fuel}} \right)$$

Liquid water (HHV)

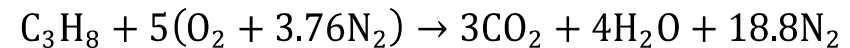
Vapour water (LHV)

- > Calculate the enthalpy of combustion at 25 °C, on both kilomol and kilogram basis, a) for liquid propane with liquid water in the products and b) for gaseous water in the products. The enthalpy of evaporation for propane is 370 kJ/kg.

Suppositions: 1) complete combustion, 2) all gases are ideal gases, 3) all system is at 25°C, 1 atm



Balanced reaction equation:



Enthalpy of combustion for the process with liquid water at products:

$$\bar{h}_C = N_{\text{CO}_2} \bar{h}_{\text{CO}_2} + N_{\text{H}_2\text{O}(l)} \bar{h}_{\text{H}_2\text{O}(l)} + N_{\text{N}_2} \bar{h}_{\text{N}_2} - (N_{\text{C}_3\text{H}_8(l)} \bar{h}_{\text{C}_3\text{H}_8(l)} + N_{\text{O}_2} \bar{h}_{\text{O}_2} + N_{\text{N}_2} \bar{h}_{\text{N}_2})$$

$$\bar{h}_C = (3)(-393520) + (4)(-285830) - (1)(-103850 - (370 \cdot 44))$$

$$\bar{h}_C = -2\,203\,750 \frac{\text{kJ}}{\text{kmol}} = -50\,085.227 \frac{\text{kJ}}{\text{kg}}$$

Since the water is liquid: $\text{HHV} = 50\,085.227 \frac{\text{kJ}}{\text{kg fuel}}$

For b) water in the products is vapour:

$$\bar{h}_C = (3)(-393520) + (4)(-241\,820) - (1)(-103850 - (370 \cdot 44))$$

$$\bar{h}_C = -2\,027\,710 \frac{\text{kJ}}{\text{kmol}} = -46\,084.318 \frac{\text{kJ}}{\text{kg}}$$

$$\text{LHV} = 46\,084.318 \frac{\text{kJ}}{\text{kg fuel}}$$



Energy balance for open systems



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Volume control

- › The energy equation for a steady-state, steady-flow process at rest is the following:

$$Q - W = \sum_{prod} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) - \sum_{react} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) \quad \left(\frac{\text{kJ}}{\text{kmol fuel}} \right)$$

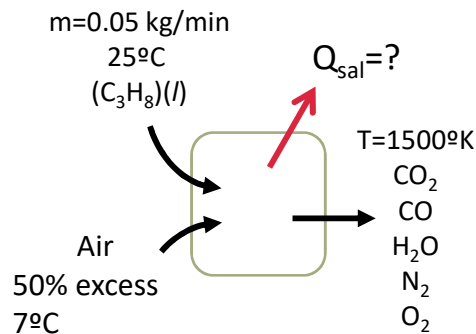
- › Must remember:
 - Heat transfer towards the system (+)
 - Work done by the system (+)
- › Normally when we examine systems where:
 - No work is done by or to the system
 - There is no heat input to the system

$$Q_{out} = \sum_{react} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) - \sum_{prod} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) \quad \left(\frac{\text{kJ}}{\text{kmol fuel}} \right)$$

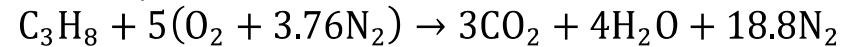
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- > Liquid propane enters a combustion chamber at 25 °C with a mass flow rate of 0.05 kg/min, where it mixes and burns with 50% excess air at 7°C. The analysis of the product gases revealed that all the hydrogen present in the fuel oxidized completely to H₂O but only the 90% of the carbon present is burned to CO₂, while the rest formed CO. If the temperature of the product gasses is 1500 K, determine a) the required air mass flow and b) the heat transfer of the combustion chamber.

Suppositions: 1) all gases are ideal gases 2) stable state, 3) Ek=Ep=0, 4) W=0, Qin=0



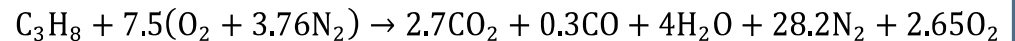
Stoichiometric balanced equation



Air excess in moles:

$$\% \text{ theoric air} = \frac{N_{\text{air real}}}{N_{\text{air stoic}}} \quad N_{\text{air real}} = (1.5)(5) = 7.5$$

Balanced equation for the real reaction



Enthalpy values in kJ/kmol

	hf	h _{280K}	h _{298K}	h _{1500K}
C ₃ H _{8(l)}	-120130	---	---	---
O ₂	0	8150	8682	49292
N ₂	0	8141	8669	47073
H ₂ O _(g)	-241820	---	9904	57999
CO ₂	-393520	---	9364	71078
CO	-110530	---	8669	47517

$$AF_{\text{stoic}} = 23.529 \frac{\text{kg}_{\text{air}}}{\text{kg}_{\text{fuel}}}$$

Determine the air flow require for the process:

$$\dot{m}_{\text{air}} = AF_{\text{stoic}} \dot{m}_{\text{fuel}} = 1.176 \frac{\text{kg}_{\text{air}}}{\text{min}}$$

› Energy balance for the combustion process steady-state:

$$Q - W = \sum_{prod} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) - \sum_{react} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) \quad \left(\frac{\text{kJ}}{\text{kmol fuel}} \right)$$

– No work transfer, no heat input

$$Q_{out} = \sum_{react} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) - \sum_{prod} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) \quad \left(\frac{\text{kJ}}{\text{kmol fuel}} \right)$$

$$\begin{aligned} Q_{out} = & (1)(-120130 + h_{298} - h_{298}) + (7.5)(0 + 8150 - 8682) \\ & + (28.2)(0 + 8141 - 8669) - (2.7)(-393520 + 71078 - 9364) \\ & - (0.3)(-110530 + 47517 - 8669) - (4)(-241820 + 57999 - 9904) \\ & - (2.65)(0 + 49292 - 8682) - (28.2)(0 + 47073 - 8669) \end{aligned}$$

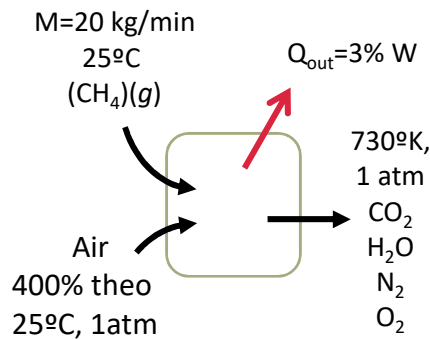
– Heat transfer for a fuel flow of 0.05 kg/min

$$Q_{out} = 362\,661.90 \frac{\text{kJ}}{\text{kmol fuel}} = 8\,242.315 \frac{\text{kJ}}{\text{kg fuel}}$$

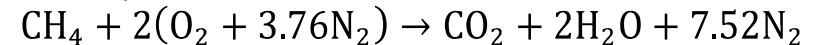
$$\dot{Q}_{out} = 412.116 \frac{\text{kJ}}{\text{min}} = 6.8686 \text{ kW}$$

- > Methane at 25°C, enters the combustor of a simple open gas turbine power plant and burns completely with 400% of theoretical air entering the compressor at 25°C, 1 atm. Products of combustion exit the turbine at 730 K, 1 atm. The rate of heat transfer from the power plant is estimated as 3% of the net power developed. Determine the net power developed, in MW, if the fuel mass flow rate is 20 kg/min.

Suppositions: 1) all gases are ideal gases 2) stable state, 3) $E_k = E_p = 0$, 4) $Q_{in} = 0$



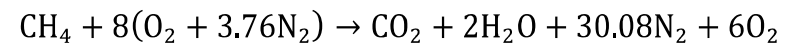
Stoichiometric balanced equation



Air excess in moles:

$$\% \text{ theoric air} = \frac{N_{\text{air real}}}{N_{\text{air stoic}}} \quad N_{\text{air real}} = (2)(4) = 8$$

Balanced equation for the real reaction



Energy balance of the system:

$$\begin{aligned}
 Q - W &= \sum_{\text{prod}} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) - \sum_{\text{react}} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) \\
 &\quad \downarrow Q = -0.03W \\
 -1.03W &= \sum_{\text{prod}} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) - \sum_{\text{react}} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) \\
 &\quad \downarrow \\
 1.03W &= \sum_{\text{react}} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ) - \sum_{\text{prod}} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ)
 \end{aligned}$$

Energy balance of the system:

$$1.03W = (1)(\bar{h}_f^\circ + \Delta h)_{CH_4} + (8)(\bar{h}_f^\circ + \Delta h)_{O_2} + (1)(\bar{h}_f^\circ + \Delta h)_{N_2} - (1)(\bar{h}_f^\circ + \Delta h)_{CO_2} \\ - (1)(\bar{h}_f^\circ + \Delta h)_{H_2O} - (1)(\bar{h}_f^\circ + \Delta h)_{O_2} - (1)(\bar{h}_f^\circ + \Delta h)_{N_2}$$

$$1.03W = (1)(-74850 - (1)(-393520 + 28662 - 9364) \\ - (2)(-241820 + 25218 - 9904) + (6)(22177 - 8682) \\ + (30.08)(21529 - 8669)$$

$$1.03W = 284\,625 \frac{\text{kJ}}{\text{kmol}} = 17\,789 \frac{\text{kJ}}{\text{kg}}$$

$$W = \frac{20 \frac{\text{kg}}{\text{min}} \cdot 17789.06 \frac{\text{kJ}}{\text{kg}} \cdot \frac{\text{min}}{60 \text{ s}}}{1.03 (1000)} = 5.7569 \text{ MW}$$

Enthalpy values in kJ/kmol

	hf	h _{298K}	h _{730K}
CH ₄	-74 850	---	---
O ₂	0	8682	22177
N ₂	0	8669	21529
H ₂ O _(g)	-241820	9904	25218
CO ₂	-393520	9364	28662

The net power developed when we burn 20 kg/min in the combustor



Energy balance closed systems



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Closed system

- › For mass constant systems in steady-state at rest we will use the following general equation:

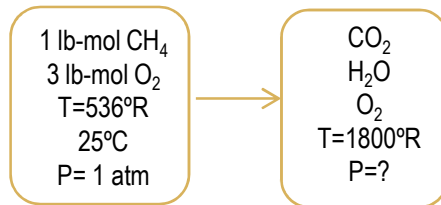
$$Q - W = \Delta(H - PV) \quad \left(\frac{\text{kJ}}{\text{kmol fuel}} \right)$$

$$Q - W = \sum_{prod} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ - P\bar{V}) - \sum_{react} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ - P\bar{V})$$

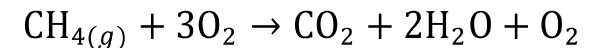
$$Q - W = \sum_{prod} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ - R_u T) - \sum_{react} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ - R_u T)$$

- › For a process where there is no work transfer and no heat applied to the system: $Q_{out} = \sum_{react} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ - R_u T) - \sum_{prod} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ - R_u T)$

- > A mixture of 1 lbmol of gaseous methane and 3 lbmol of oxygen is contained in a constant volume container at 77 °F and 1 atm. The mix is ignited and the methane gas burns completely. If the final temperature of the system is 1800 R, determine a) the final pressure of the container and b) the amount of heat transfer.



Stoichiometric balanced equation



Energy balance for closed system:

$$Q_{out} = \sum_{react} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ - R_U T) - \sum_{prod} N_i (\bar{h}_f^\circ + h - \bar{h}^\circ - R_U T)$$

$$\begin{aligned} Q_{out} = & (1)(-32210 - 1.986 \cdot 537) + (3)(-1.986 \cdot 537) \\ & - (1)(-169300 + 18391.5 - 4027.5 - 1.986 \cdot 1800) \\ & - (2)(-104040 + 15433 - 4258 - 1.986 \cdot 1800) - (1)(13485.8 \\ & - 3725.1 - 1.986 \cdot 1800) \end{aligned}$$

Enthalpy values in BTU/kmol

	hf	h _{537R}	h _{1800R}
CH _{4(g)}	-32210	---	---
O ₂	0	3725.1	13485.1
H ₂ O _(g)	-169300	4258.0	15433.0
CO ₂	-104040	4027.5	18391.5

$$Q_{out} = 308730 \frac{\text{Btu}}{\text{lbmol}} = 19300 \frac{\text{Btu}}{\text{kg}}$$

Final pressure of the container:

$$p_{prod} = p_{react} \left(\frac{N_{prod}}{N_{react}} \right) \left(\frac{T_{prod}}{T_{react}} \right) = 3.35 \text{ atm}$$